1

```
clf
v=[-3;-3;8;9]
```

```
v = 4×1
    -3
    -3
     8
     9
```

```
A=[3,1,-1;1,2,0;0,1,2;1,1,-1]
```

```
A = 4×3
     3     1    -1
     1     2     0
     0     1     2
     1     1    -1
```

```
n=rank(A)
```

```
n = 
3
```

```
p1=A*pinv(A)*v
```

```
p1 = 4×1
   -2.0000
    3.0000
    4.0000
         0
```

```
p2=A*inv((A'*A))*A'*v
```

```
p2 = 4×1
   -2.0000
    3.0000
    4.0000
   -0.0000
```

2

```
clf
v=[2;1;3]
```

```
v = 3×1
     2
     1
     3
```

```
A=[1,1;1,2;0,1]
```

```
A = 3×2
     1     1
     1     2
     0     1
```

```
n=rank(A)
```

```
n =
2
```

```
p1=A*pinv(A)*v
```

```
p1 = 3×1
    0.6667
    2.3333
    1.6667
```

```
p2=A*inv((A'*A))*A'*v
```

```
p2 = 3×1
    0.6667
    2.3333
    1.6667
```

3

```
clf
v=[1;3;9]
```

```
v = 3×1
     1
     3
     9
```

```
A=[1,1,2,4;1,2,3,6;0,1,1,2]
```

```
A = 3×4
     1     1     2     4
     1     2     3     6
     0     1     1     2
```

```
n=rank(A)
```

```
n =
2
```

```
p1=A*pinv(A)*v
```

```
p1 = 3×1
   -1.3333
    5.3333
    6.6667
```

4

```
clf
v = [1; -2; 8; 1]
```

```
v = 4×1
     1
    -2
     8
     1
```

```
a = [1; 2; 3; -1]
```

```
a = 4×1
     1
     2
     3
    -1
```

```
p = a*((a'*v)/(a'*a))
```

```
p = 4×1
    1.3333
    2.6667
    4.0000
   -1.3333
```

5

```
clf
v = [2;3;7]
```

```
v = 3×1
     2
     3
     7
```

```
a1=[2;-1;1]
```

```
a1 = 3×1
     2
    -1
     1
```

```
a2=[1;-1;3]
```

```
a2 = 3×1
     1
    -1
     3
```

```
a = cross(a1, a2)
```

```
a = 3×1
    -2
    -5
    -1
```

```
p = a*((a'*v)/(a'*a))
```

```
p = 3×1
    1.7333
    4.3333
    0.8667
```

6

```
clf
v = [1;10;30]
```

```
v = 3×1
     1
    10
    30
```

```
a = [1; 2; 3]
```

```
a = 3×1
     1
     2
     3
```

```
p = a*((a'*v)/(a'*a))
```

```
p = 3×1
    7.9286
   15.8571
   23.7857
```

Regression using matlab

```
clf
[B,text]=xlsread('C:\Users\Hp\OneDrive\Documents\xlformatlabReg1.xlsx');
x=B(:,1);
y=B(:,2);
n=length(x);
A=[x,ones(n,1)];
ATA=[sum(x.^2),sum(x);sum(x),n]
```

```
ATA = 2×2
   770     0
     0    21
```

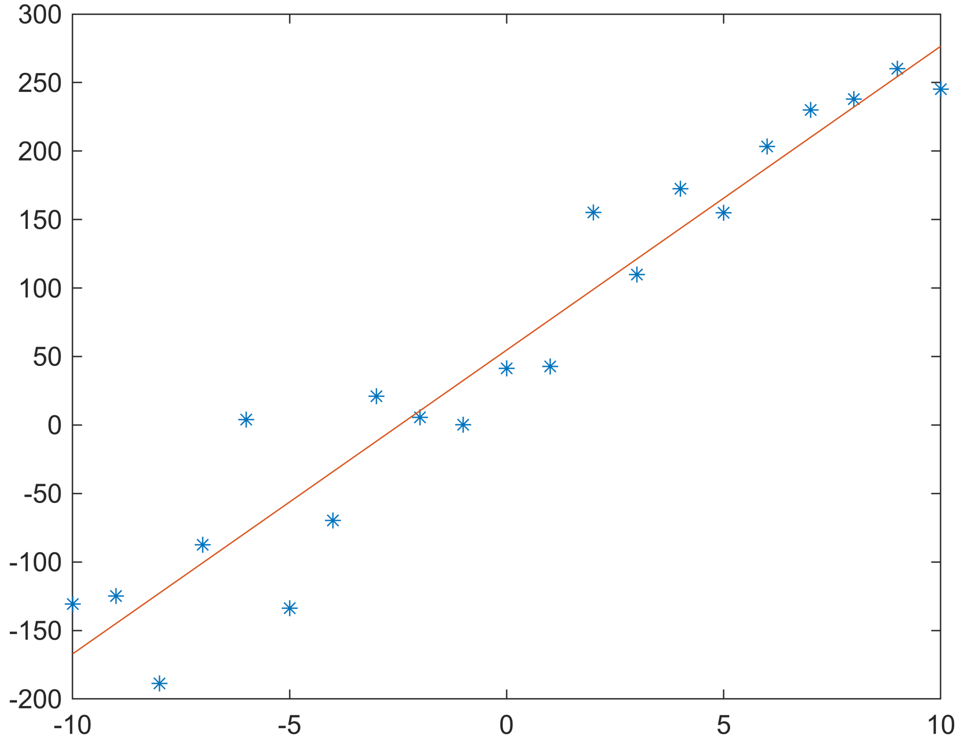
```
ATb=[sum(x.*y);sum(y)]
```

```
ATb = 2×1
10^4 ×
    1.7079
    0.1149
```

```
m_and_c=inv(ATA)*ATb
```

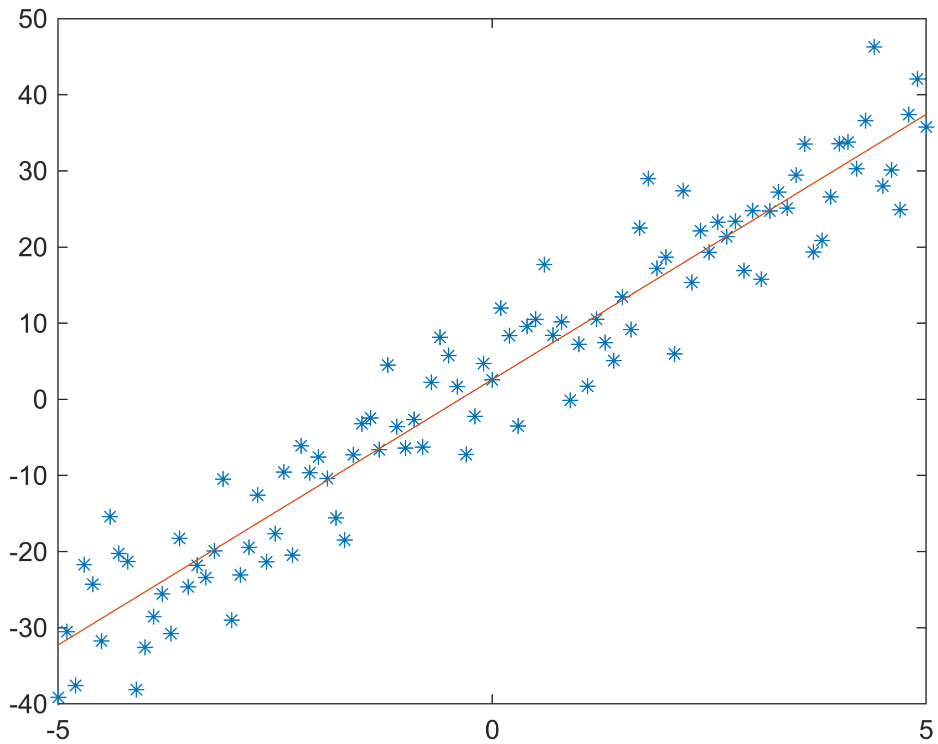
```
m_and_c = 2×1
   22.1803
   54.7134
```

```
y_projected=A*inv(ATA)*ATb;
plot(x,y,'*');
hold on
plot(x,y_projected);
```



linear regression generation of data

```
clf
m=7;
c=3;
x=-5:0.1:5;                    % a row vector x
n=length(x);
y=m*x+c;
noise_stdev=6;
                               % noise standard deviation
noise=noise_stdev*randn(1,n);
yn=y+noise;
plot(x,yn,'*');
                               % plot of generated points
hold on
                               % fitting a line or re-estimating m and c
                               % create matrix A with x and ones
A=[x' ones(n,1)];
yn=yn';                        % make yn a column vector
m_and_c=pinv(A)*yn;            % estimate m and c
y_projected=A*pinv(A)*yn;      % y estimated
plot(x,y_projected);
```




```
m_and_c
```

```
m_and_c = 2×1
    6.9687
    2.5836
```

Estimation of m and c

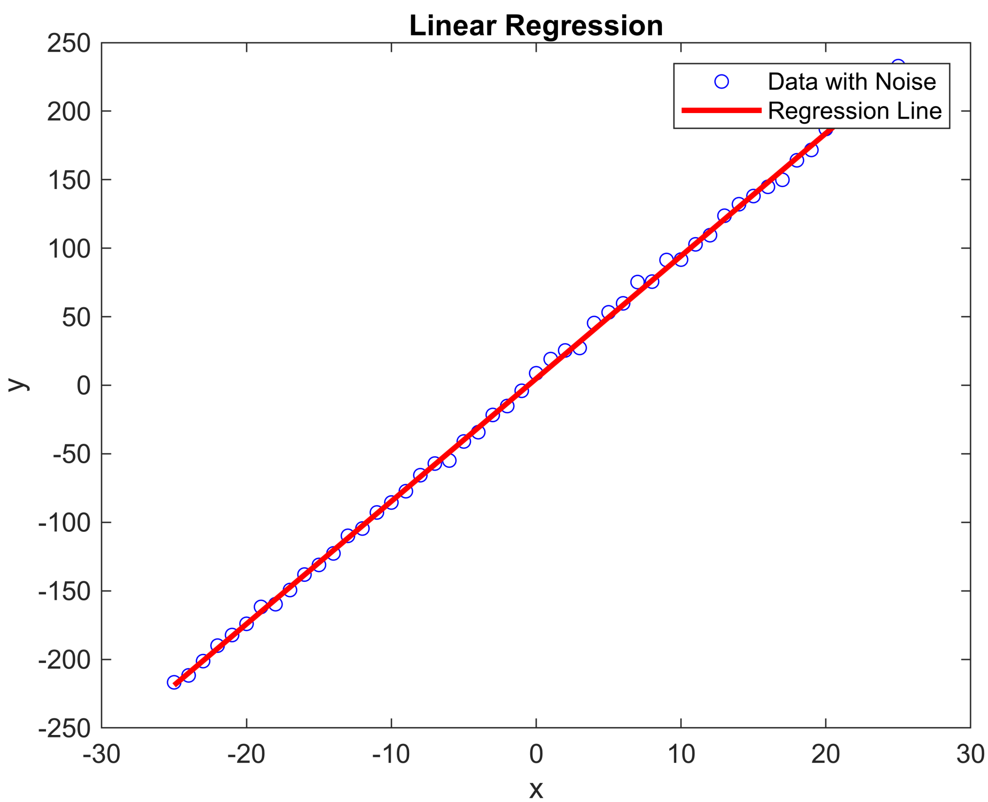
```
% stem 1: Generate the data for regression with noise
m = 9;
c = 5;
x = (-25:25)';
y_true = m * x + c;
noise_std = 3;
noise = noise_std * randn(size(x));
y = y_true + noise;

% Step 2: Perform linear regression to estimate m and c
A = [x, ones(size(x))];
params = A\y; % Parameters estimation using least squares
m_estimated = params(1)
```

```
m_estimated =
8.9494
```

```
c_estimated = params(2)
```

```
c_estimated =
4.8498
```

```
% Step 3: Plot the original data and regression line
figure;
plot(x, y, 'bo', 'MarkerSize', 5); % Original data points
hold on;
plot(x, m_estimated*x + c_estimated, 'r-', 'LineWidth', 2); % Regression line
hold off;
xlabel('x');
ylabel('y');
title('Linear Regression');
legend('Data with Noise', 'Regression Line');
```



8b

```
clf
[B,text]=xlsread('C:\Users\Hp\OneDrive\Documents\xlformatReg2.xlsx');
x=B(:,1);
y=B(:,2);
n=length(x);
A=[x,ones(n,1)];
ATA=[sum(x.^2),sum(x);sum(x),n]
```

```
ATA = 2×2
```

```
  192.5000         0
         0   21.0000
```

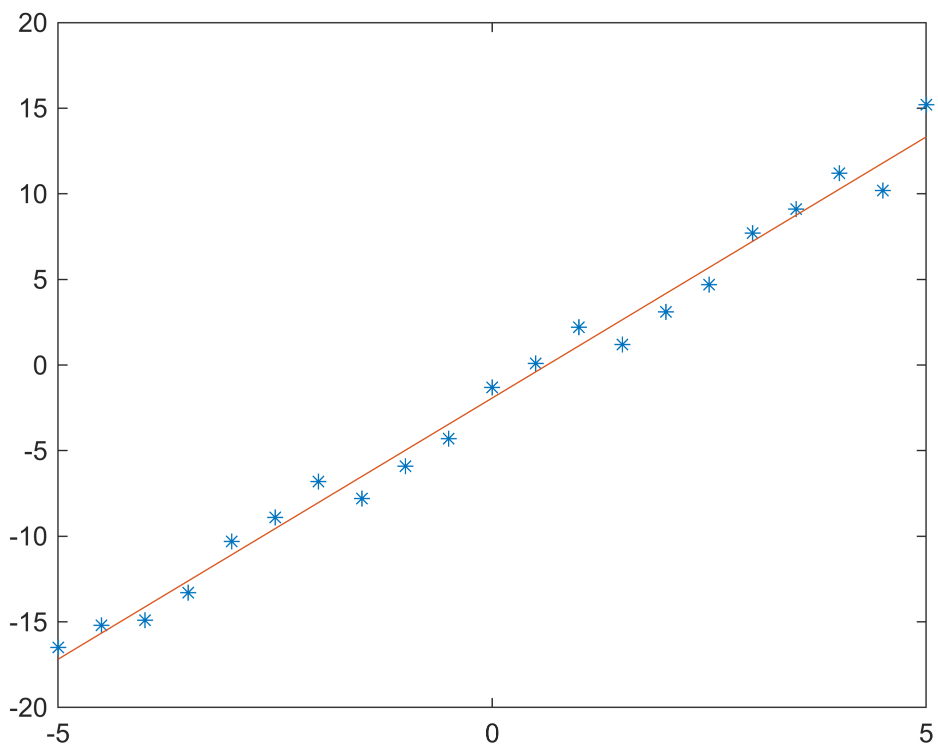
```
ATb=[sum(x.*y);sum(y)]
```

```
ATb = 2×1
  587.2000
  -40.5000
```

```
m_and_c=inv(ATA)*ATb
```

```
m_and_c = 2×1
    3.0504
   -1.9286
```

```
y_projected=A*inv(ATA)*ATb;
plot(x,y,'*');
hold on
plot(x,y_projected);
```

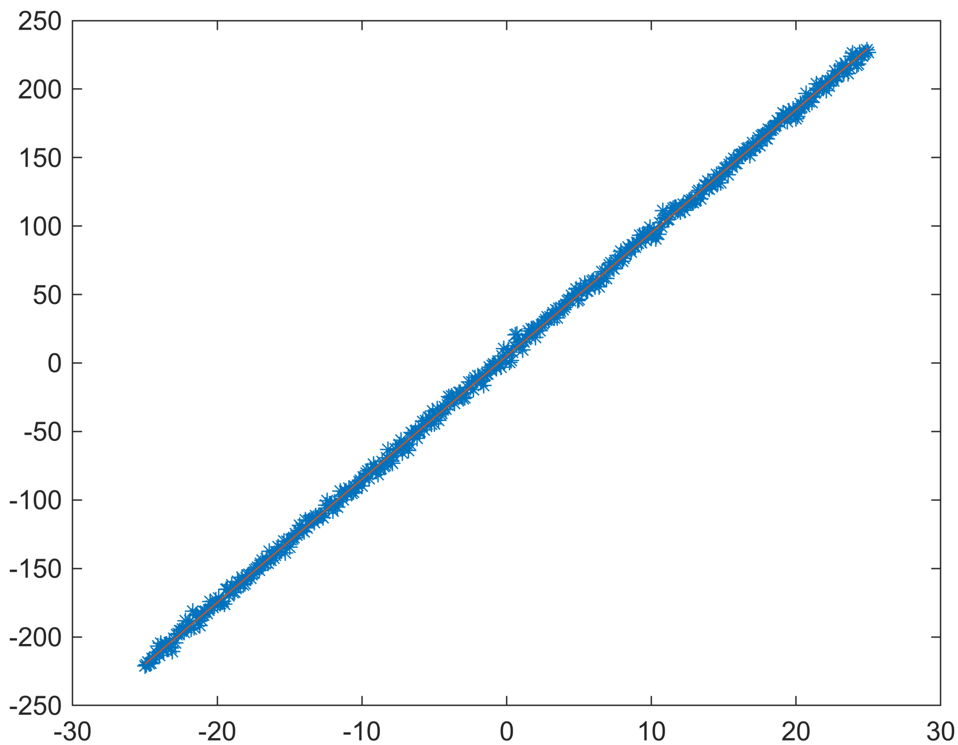


practice questions

1

```
clf
m=9;
c=5;
x=-25:0.1:25;
```

```
n=length(x);
y=m*x+c;
noise_stdev=3;
noise=noise_stdev*randn(1,n);
yn=y+noise;
plot(x,yn,'*');
hold on
A=[x' ones(n,1)];
yn=yn';
m_and_c=pinv(A)*yn;
y_projected=A*pinv(A)*yn;
plot(x,y_projected);
```




```
m_and_c
```

```
m_and_c = 2×1
    8.9945
    5.0903
```

2

```
clf
m = 9;
c = 5;
x = (-25:25)';
y_true = m * x + c;
noise_std = 3;
```

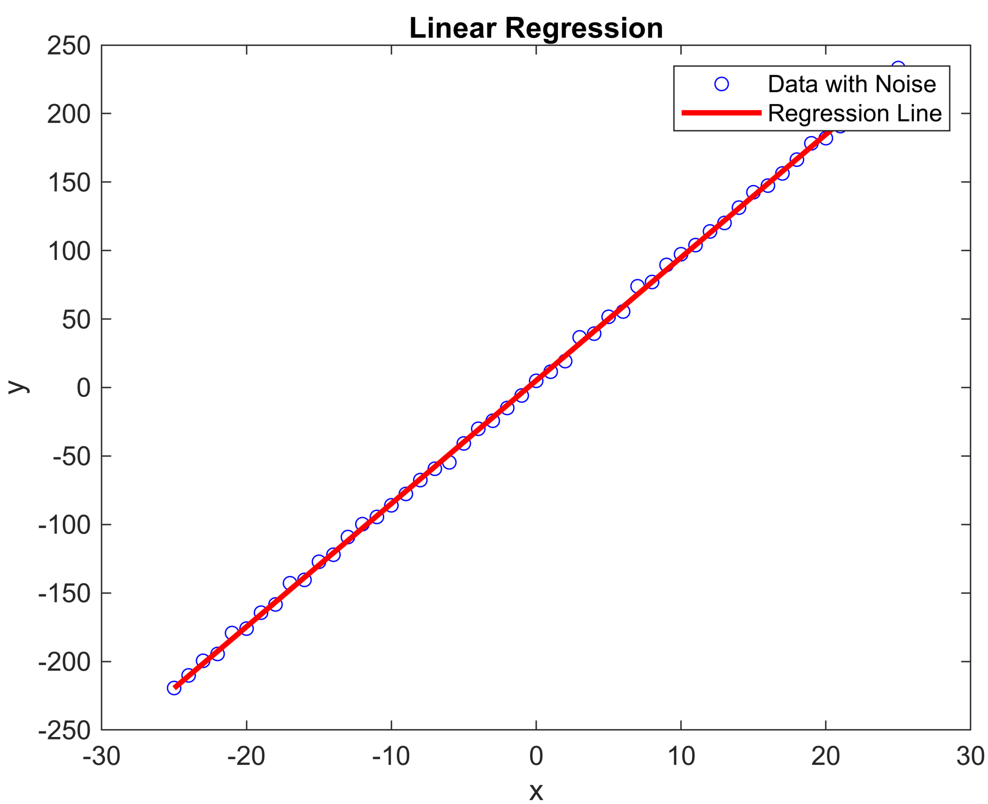
```
noise = noise_std * randn(size(x));
y = y_true + noise;
A = [x, ones(size(x))];
params = A\y;
m_estimated = params(1)
```

```
m_estimated =
8.9850
```

```
c_estimated = params(2)
```

```
c_estimated =
5.1115
```

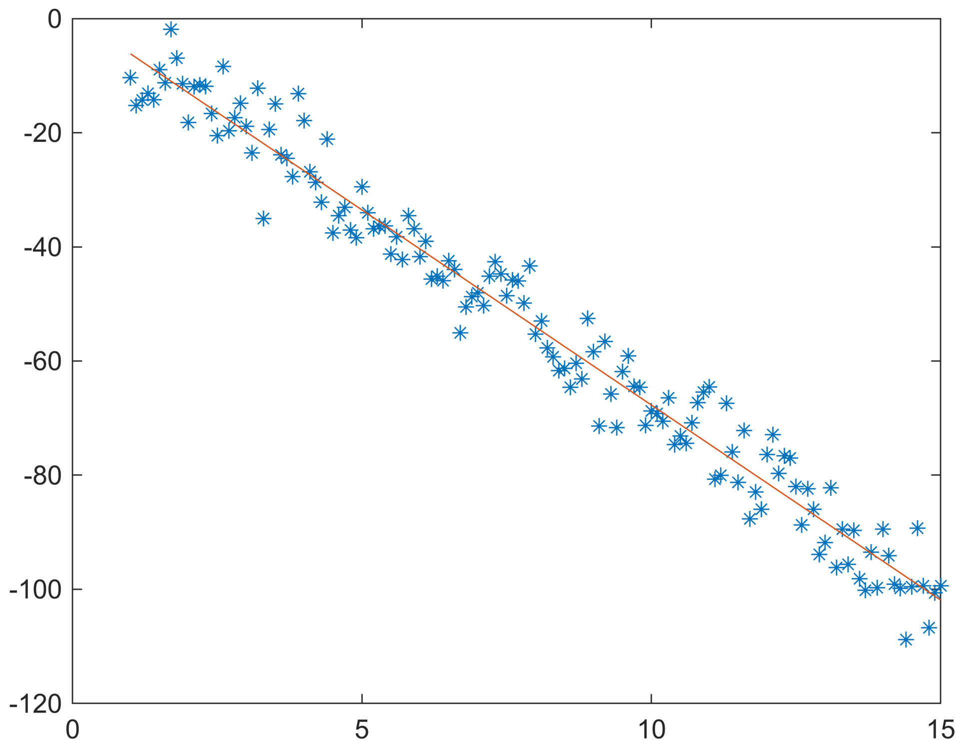
```
figure;
plot(x, y, 'bo', 'MarkerSize', 5);
hold on;
plot(x, m_estimated*x + c_estimated, 'r-', 'LineWidth', 2);
hold off;
xlabel('x');
ylabel('y');
title('Linear Regression');
legend('Data with Noise', 'Regression Line');
```



3

```
clf
m=-7;
```

```
c=2;
x=1:0.1:15;
n=length(x);
y=m*x+c;
noise_stdev=5;
noise=noise_stdev*randn(1,n);
yn=y+noise;
plot(x,yn,'*');
hold on
A=[x' ones(n,1)];
yn=yn';
m_and_c=pinv(A)*yn;
y_projected=A*pinv(A)*yn;
plot(x,y_projected);
```




```
m_and_c
```

```
m_and_c = 2×1
   -6.8368
    0.6568
```

4

```
clf
m = -7;
c = 2;
x = (1:15)';
```

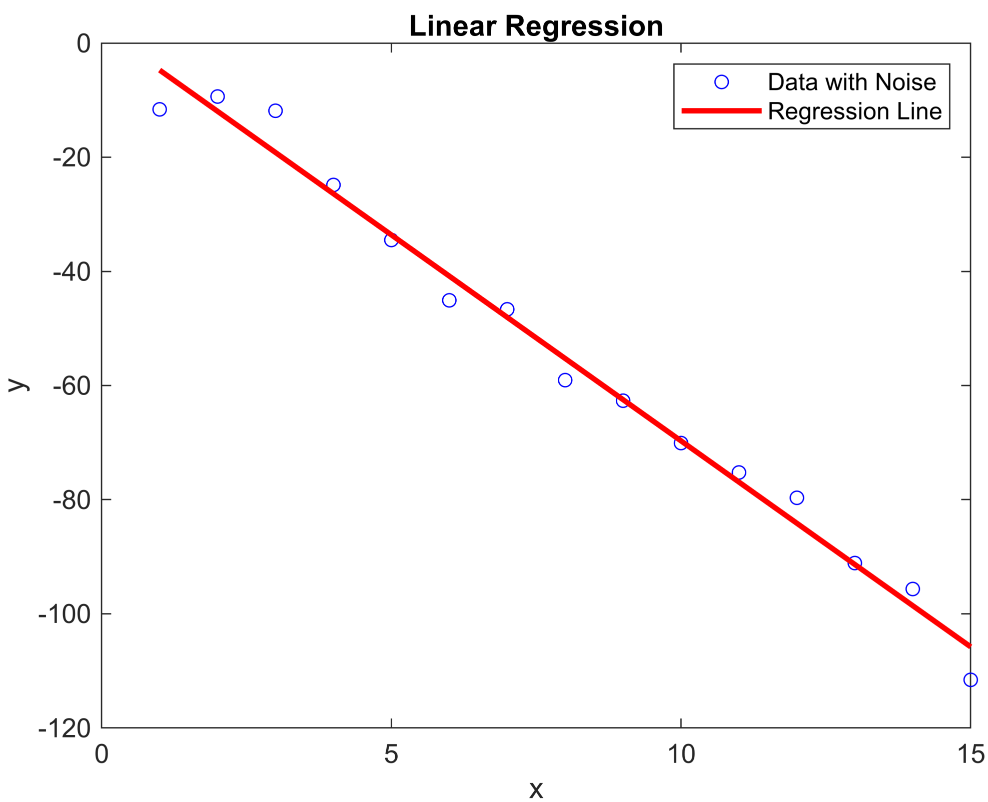
```
y_true = m * x + c;
noise_std = 5;
noise = noise_std * randn(size(x));
y = y_true + noise;
A = [x, ones(size(x))];
params = A\y;
m_estimated = params(1)
```

```
m_estimated =
-7.2183
```

```
c_estimated = params(2)
```

```
c_estimated =
2.4826
```

```
figure;
plot(x, y, 'bo', 'MarkerSize', 5);
hold on;
plot(x, m_estimated*x + c_estimated, 'r-', 'LineWidth', 2);
hold off;
xlabel('x');
ylabel('y');
title('Linear Regression');
legend('Data with Noise', 'Regression Line');
```



5a

```
clf
A=[1,1;2,1;3,1;4,1;5,1]
```

```
A = 5×2
     1     1
     2     1
     3     1
     4     1
     5     1
```

```
b=[10;10;16;27;23]
```

```
b = 5×1
    10
    10
    16
    27
    23
```

```
n=rank(A)
```

```
n = 
2
```

```
x1=pinv(A)*b
```

```
x1 = 2×1
    4.3000
    4.3000
```

```
x2=inv((A'*A))*A'*b
```

```
x2 = 2×1
    4.3000
    4.3000
```

5b

```
clf
A=[2,-3,1;1,3,-4;3,-2,-1;5,-1,-4]
```

```
A = 4×3
     2    -3     1
     1     3    -4
     3    -2    -1
     5    -1    -4
```

```
b=[1;-4;-1;-4]
```

```
b = 4×1
     1
    -4
    -1
    -4
```

```
n=rank(A)
```

```
n = 
2
```

```
x1=pinv(A)*b
```

```
x1 = 3×1
   -0.3333
   -0.3333
    0.6667
```

5c

```
clf
A=[2,1;2,-3;3,1;4,1]
```

```
A = 4×2
     2     1
     2    -3
     3     1
     4     1
```

```
b=[20;4;28;36]
```

```
b = 4×1
    20
     4
    28
    36
```

```
n=rank(A)
```

```
n =
2
```

```
x1=pinv(A)*b
```

```
x1 = 2×1
    8.0000
    4.0000
```

```
x2=inv((A'*A))*A'*b
```

```
x2 = 2×1
     8
     4
```

5d

```
clf
A=[1,3,-5;2,-3,8;4,3,-2;9,-3,1;3,-5,3]
```

```
A = 5×3
     1     3    -5
     2    -3     8
     4     3    -2
     9    -3     1
     3    -5     3
```

```
b=[-1;7;5;8;0]
```

```
b = 5×1
    -1
     7
     5
     8
     0
```

```
n=rank(A)
```

```
n =
3
```

```
x1=pinv(A)*b
```

```
x1 = 3×1
    1.0802
    1.1065
    1.0294
```

```
x2=inv((A'*A))*A'*b
```

```
x2 = 3×1
    1.0802
    1.1065
    1.0294
```